

**ENTERPRISE AI KNOWLEDGE ASSISTANT***Enterprise Project Management Toolkit*

# Functional Requirements Document

FRD – Enterprise AI Knowledge Assistant

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|-------------------------|-------------------------------------------|
| <b>Document Title</b>   | Functional Requirements Document (FRD)    |
| <b>Project Name</b>     | Enterprise AI Knowledge Assistant (EAIKA) |
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| <b>Product Owner</b>    | Mautik Patel – Product Owner              |
| <b>Engineering Lead</b> | Michael Chen                              |
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| <b>Status</b>           | Approved                                  |
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## 1. Purpose

This Functional Requirements Document (FRD) translates the business requirements defined in the BRD into specific, testable system functions. It describes what the Enterprise AI Knowledge Assistant (EAIKA) must do, organized by functional module: Document Ingestion, Knowledge Base Management, Semantic Search & Retrieval, Answer Generation, Source Transparency, End-User Interface, and Administrator Dashboard.

## 2. Functional Modules Overview

| Module                  | Description                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------|
| Document Ingestion      | Upload and processing of enterprise documents into text-extractable, chunked content.         |
| Embedding & Indexing    | Conversion of document chunks into vector embeddings and storage in ChromaDB.                 |
| Semantic Retrieval      | Similarity-based search across the vector store to find the most relevant chunks for a query. |
| Answer Generation (LLM) | Local LLM (Qwen2.5 via Ollama) synthesizes a natural-language answer from retrieved chunks.   |
| Source Transparency     | Presentation of source document references alongside every generated answer.                  |
| End-User Chat Interface | Streamlit-based conversational UI for asking questions and viewing answers/sources.           |
| Administrator Dashboard | Document upload, knowledge base build/rebuild, and inventory/statistics management.           |

## 3. Functional Requirements

### 3.1 Document Ingestion

| ID    | Requirement                                                                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-01 | The system shall accept document uploads in PDF, DOCX, PPTX, XLSX, TXT, CSV, EML, PNG, JPG, and JPEG formats.                                             |
| FR-02 | The system shall extract text content from each supported format, including OCR-based text extraction from image files and scanned/image-based PDF pages. |
| FR-03 | The system shall split extracted text into overlapping chunks sized appropriately for embedding and retrieval accuracy.                                   |
| FR-04 | The system shall reject or flag files that fail extraction and report the failure to the administrator.                                                   |

### 3.2 Embedding & Indexing

| ID    | Requirement                                                                                                                             |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------|
| FR-05 | The system shall generate vector embeddings for each document chunk using the configured embedding model.                               |
| FR-06 | The system shall store embeddings, chunk text, and source metadata (document name, location, page/section) in ChromaDB.                 |
| FR-07 | The system shall support a full knowledge base “build” operation that processes all newly uploaded documents.                           |
| FR-08 | The system shall support a “rebuild” operation that re-indexes the entire vector database, for example after an embedding model change. |

### 3.3 Semantic Retrieval

| ID    | Requirement                                                                                                                                                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-09 | The system shall accept a natural-language question from the user and convert it into a query embedding.                                                      |
| FR-10 | The system shall retrieve the top-N most semantically similar document chunks from ChromaDB for a given query.                                                |
| FR-11 | The system shall return relevant results even when the user's wording differs from the source document's exact terminology (semantic, not keyword, matching). |

### 3.4 Answer Generation

| ID    | Requirement                                                                                                                                    |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-12 | The system shall pass retrieved chunks and the user's question to the local LLM (Qwen2.5 via Ollama) to generate a natural-language answer.    |
| FR-13 | The system shall constrain the LLM's answer to information contained in the retrieved context wherever possible, to reduce unsupported claims. |
| FR-14 | The system shall handle the case where no relevant document content is found and inform the user rather than fabricating an answer.            |

### 3.5 Source Transparency

| ID    | Requirement                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------------------|
| FR-15 | The system shall display the source document name(s) used to generate each answer.                              |
| FR-16 | The system shall allow the user to view or navigate to the specific retrieved passage(s) supporting the answer. |

### 3.6 End-User Chat Interface

| ID    | Requirement                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------|
| FR-17 | The system shall provide a Streamlit-based chat interface where users can type questions and view answers in a conversational format. |
| FR-18 | The system shall preserve conversation history within a user session.                                                                 |
| FR-19 | The system shall present a clear indication when the system is retrieving/generating an answer (loading state).                       |

### 3.7 Administrator Dashboard

| ID    | Requirement                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------|
| FR-20 | The system shall provide an administrator view for uploading new documents into the knowledge base.                          |
| FR-21 | The system shall display knowledge base statistics, including document count, chunk count, and last build/rebuild timestamp. |
| FR-22 | The system shall allow an administrator to trigger a knowledge base build or full rebuild on demand.                         |
| FR-23 | The system shall display a document inventory list, including file name, format, upload date, and indexing status.           |

## 4. Functional Flow Summary

End-to-end functional flow: (1) Administrator uploads documents → (2) System extracts text and performs OCR where required → (3) System chunks text and generates embeddings → (4) Embeddings and metadata are indexed in ChromaDB → (5) End user submits a natural-language question via the chat interface → (6) System retrieves the most relevant chunks → (7) Local LLM generates an answer grounded in retrieved content → (8) System returns the answer with source citations to the user.

## 5. Traceability to Business Requirements

| Business Requirement                     | Related Functional Requirements                                   |
|------------------------------------------|-------------------------------------------------------------------|
| BR-01 – Natural-language Q&A             | FR-09, FR-10, FR-12, FR-17                                        |
| BR-02 – Source traceability              | FR-06, FR-15, FR-16                                               |
| BR-03 – Local-only processing            | FR-05, FR-06, FR-12 (implemented via local Ollama/ChromaDB stack) |
| BR-04 – multi-format support             | FR-01, FR-02                                                      |
| BR-05 – Self-service document management | FR-20, FR-22                                                      |
| BR-07 – Knowledge base monitoring        | FR-21, FR-23                                                      |

## 6. Approval

This FRD is subject to review and approval by the Product Owner and Engineering Lead prior to design and development.

| Name / Role                          | Approval | Date         |
|--------------------------------------|----------|--------------|
| Priya Nataraj – Head of Data Science | Approved | Feb 20, 2026 |
| Mautik Patel – Product Owner         | Approved | Feb 21, 2026 |
| Michael Chen – Engineering Lead      | Approved | Feb 21, 2026 |